

URVISH BABARIYA

B.Tech – Information and Communication Technology

📞 : +91-8799054121 • ✉ : urvishbabariya.work.07@gmail.com

🌐 : urvish-babariya • 🔄 : Urvish2007 • 📁 : Portfolio

Education

Dhirubhai Ambani University

B.Tech in Information and Communication Technology; CPI: **7.53/10**

Gaytri Vidhya Mandir (GHSEB), 12th Grade

Percentage: **98** (99.81 PR); GUJCET: **116.25/120** (99.82 PR)

Uma Vidhyalya (GSEB), 10th Grade

Percentage: **94.50** (99.72 PR)

2024 – Present

Gandhinagar, Gujarat

2023 – 2024

Ahmedabad, Gujarat

2021 – 2022

Ahmedabad, Gujarat

Relevant Coursework

- Data Structures
- Database Management Systems
- Software Engineering
- Design & Analysis of Algorithms
- Object-Oriented Programming (OOP)
- Computer Systems Programming
- Operating Systems

Projects

PharmaChain | *React, Spring Boot 3, PostgreSQL, Spring AI, Docker*

GitHub

- Engineered a full-stack pharmaceutical supply chain platform using **React/TypeScript** and a **Spring Boot 3 REST API**, leveraging **PostgreSQL** triggers to enforce strict FDA-style drug compliance.
- Secured endpoints via **JWT authentication** (5-role RBAC) with compliant e-signatures; engineered an append-only **SHA-256 cryptographic audit ledger** to ensure immutable record-keeping.
- Developed a **Spring AI compliance copilot** combining **RAG** (via **pgvector** & **Ollama**) for regulatory documents with **Groq LLM tool-calling** to enable natural language querying of live inventory data.
- Integrated **IoT cold-chain telemetry** for real-time monitoring, and validated database logic using **Testcontainers** with CI/CD pipelines managed via **GitHub Actions** and **Docker**.

LatencyBook | *C++20, Lock-Free Concurrency, OS Internals*

GitHub

- Engineered a low-latency limit order book and alpha-signal engine in **C++20**, replacing **malloc**/mutex-based structures with a custom **pool allocator** and **lock-free SPSC ring buffer**, cutting allocation cost by **5.6×** (9.81ns → 1.76ns) and queue push/pop latency by **36.6×** (38.4ns → 1.05ns).
- Achieved a fully **allocation-free** hot path (machine-verified) with **O(1)** order-book operations (21.5ns add/cancel cycle) and **11.8×** faster memory access via **2MB huge pages**, driving a full pipeline tick-to-trade latency of **312ns (p50)**; validated by 40/40 tests under ThreadSanitizer and ASan/UBSan in CI.

CodeMentor-PRO | *Python, LangChain, Groq, Gradio*

GitHub | Demo

- Built an AI code explainer that adapts to skill level (beginner/intermediate/expert) using **Groq-hosted LLMs** via **LangChain** for structured, line-by-line reasoning.
- Deployed an interactive **Gradio** interface on **Hugging Face Spaces** for real-time, adjustable-depth explanations.

Technical Skills

Languages: Java, C, C++/C++20, SQL, Python

Backend & Frameworks: Spring Boot, Spring Security, Spring Data JPA, Hibernate, RESTful APIs, Microservices

AI & Data: Spring AI, LangChain, LangGraph, Gradio, RAG Pipelines, AI Agents (ReAct)

Low-Latency & Systems: Lock-Free Concurrency, Multithreading, Memory/Pool Allocators, OS Internals (Huge Pages, CPU Scheduling, Thread Affinity), Google Benchmark

DevOps, Tools & Testing: Git, Docker, GitHub Actions (CI/CD), CMake, Maven, JWT, Postman, JUnit, GoogleTest, Testcontainers, ThreadSanitizer/AddressSanitizer

Databases: PostgreSQL, MySQL, pgvector, ChromaDB (Vector DB)

Soft Skills: Team Collaboration, Leadership, Problem Solving, Communication

Leadership / Extracurricular

Campus Ambassador | *GeeksforGeeks x DAIICT*

Jan 2026 – Jun 2026

- Organized guidance sessions to boost platform awareness and student engagement.

Social Work Intern | *Shree Krishnam Foundation*

Dec 2025

- Mentored rural students in Mathematics and English; coordinated community health surveys and medical camps.

Club Member | *Electronics Hobby Club*

Apr 2025 – Present

- Led hands-on workshops on foundational electronics, embedded systems, and Arduino projects.

Coordinator | *Synapse'25*

Feb 2025

- Directed core event logistics: guest relations, stage operations, and cross-team communication.

Achievements

- **Codeforces: Pupil** (Max 1360) [[_vinishhhh_7](#)]
- **CodeChef: 3-Star** (Max 1612) [[_silentub_34](#)]
- **LeetCode: Knight** (Max 1956) [[_Urvish.Babariya](#)]
- Solved **1800+** coding problems [[Profile](#)]
- Secured **1st Runner-Up** in Code C Blind 2024 and Code Relay 2025
- **GUJCET/BOARD 2024: 99.82 Percentile** [Top **0.18%** of 1.37 lakh students]
- Secured All Gujarat **Rank 50** in 12th Science
- **2nd Runner-Up**, Microsoft Student Technical Club Gen-AI Hackathon